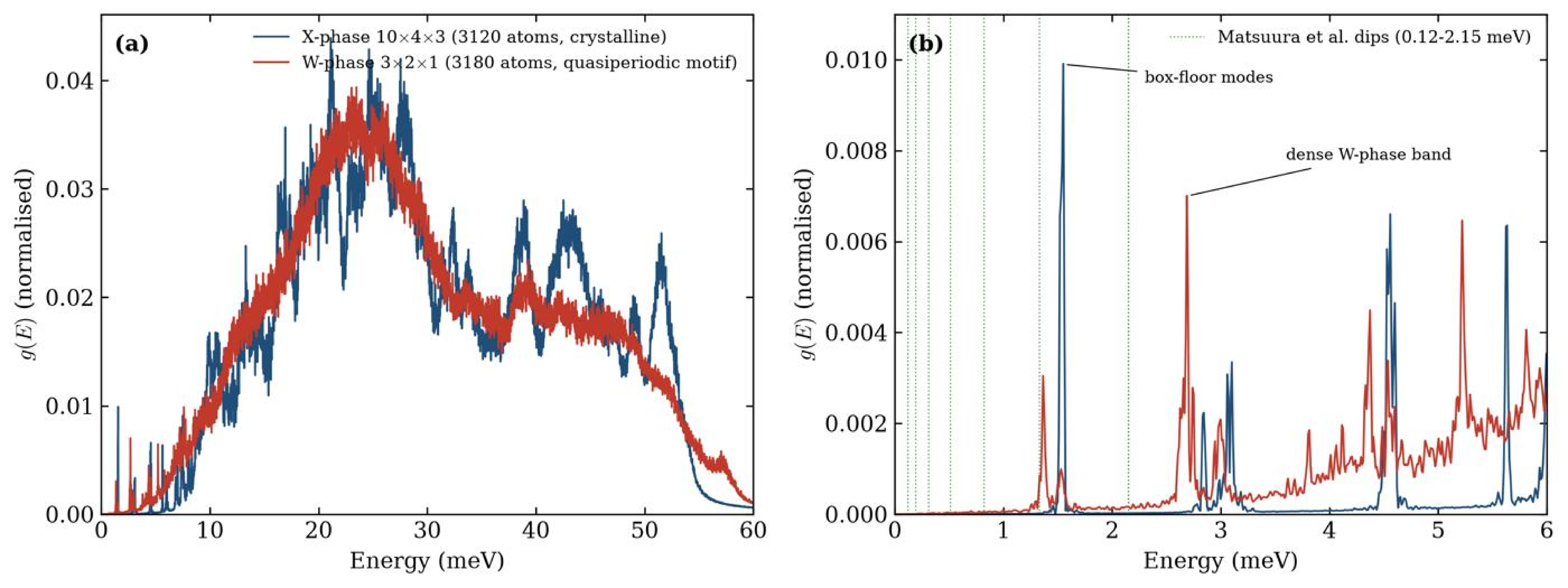
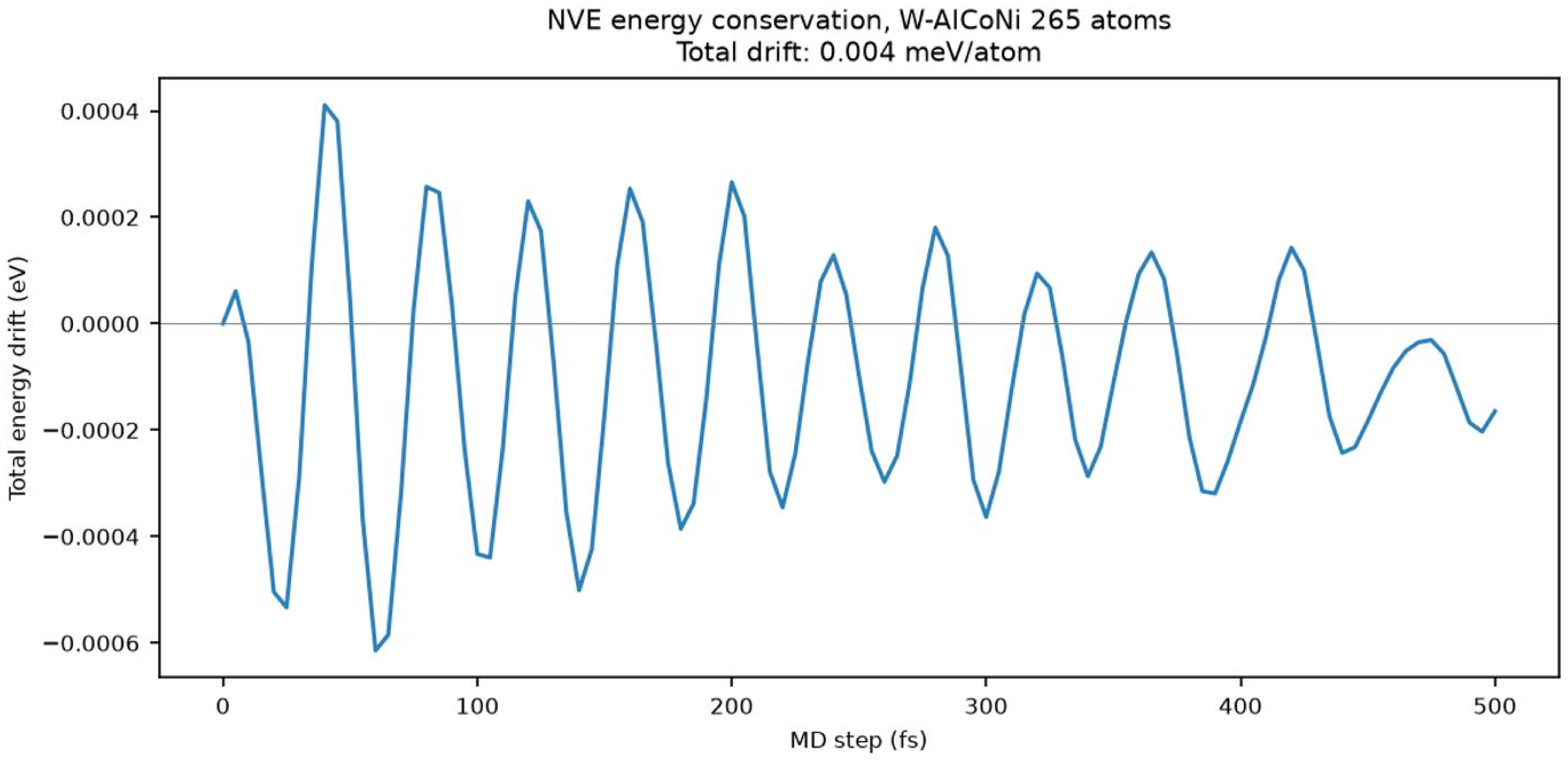
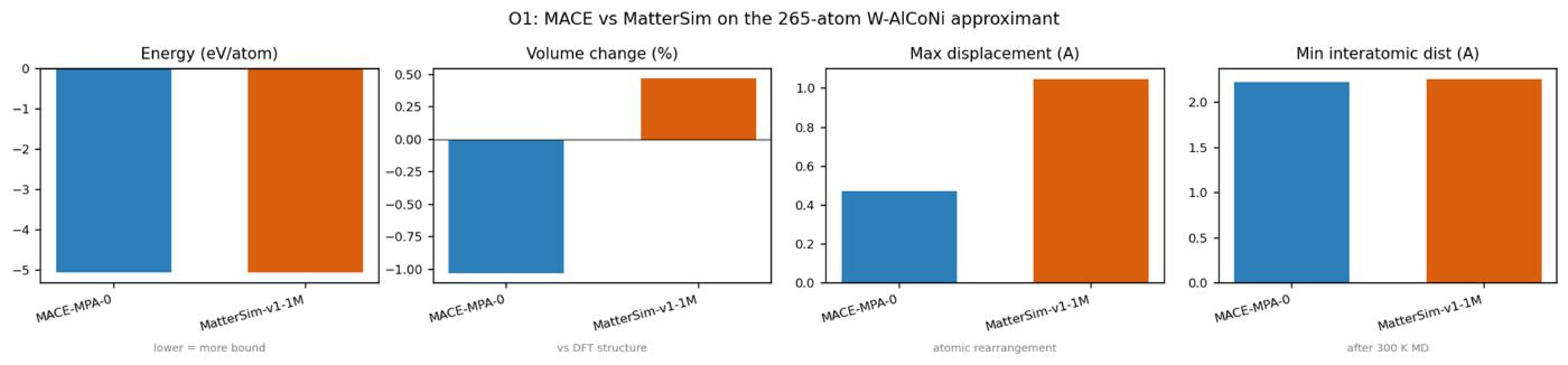
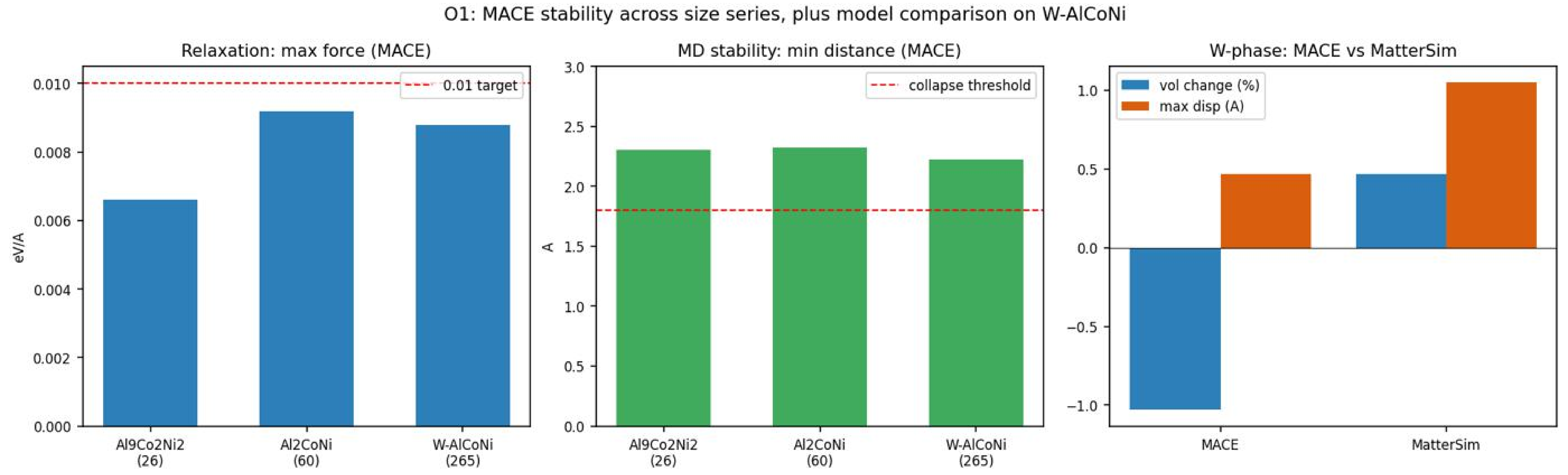


(a)
X-phase 10×4×3 (3120 atoms, crystalline)
W-phase 3×2×1 (3180 atoms, quasiperiodic motif)
g(E) (normalised)
Energy (meV)
0.00
0.01
0.02
0.03
0.04
0
10
20
30
40
50
60
(b)
Matsuura et al. dips (0.12-2.15 meV)
box-floor modes
dense W-phase band
g(E) (normalised)
Energy (meV)
0.000
0.002
0.004
0.006
0.008
0.010
0
1
2
3
4
5
6

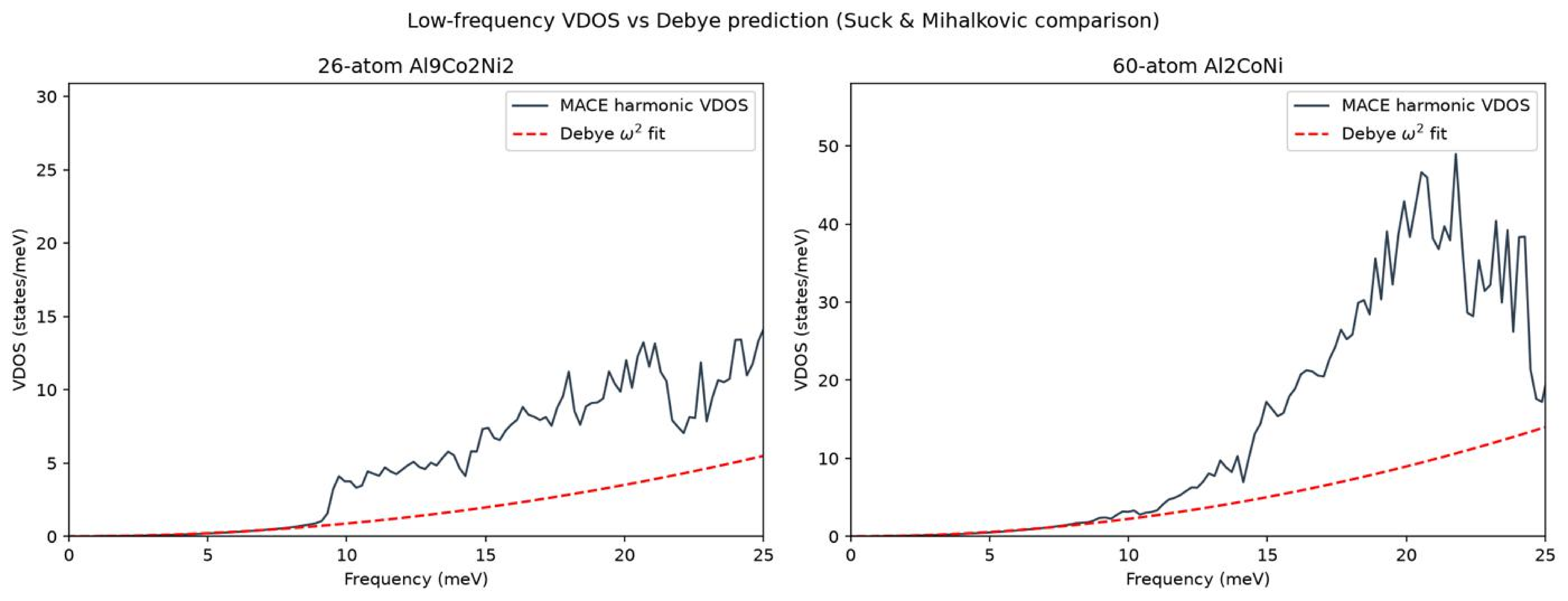

NVE energy conservation, W-AlCoNi 265 atoms
Total drift: 0.004 meV/atom
Total energy drift (eV)
0.0004
0.0002
0.0000
−0.0002
−0.0004
−0.0006
0
100
200
300
400
500
MD step (fs)



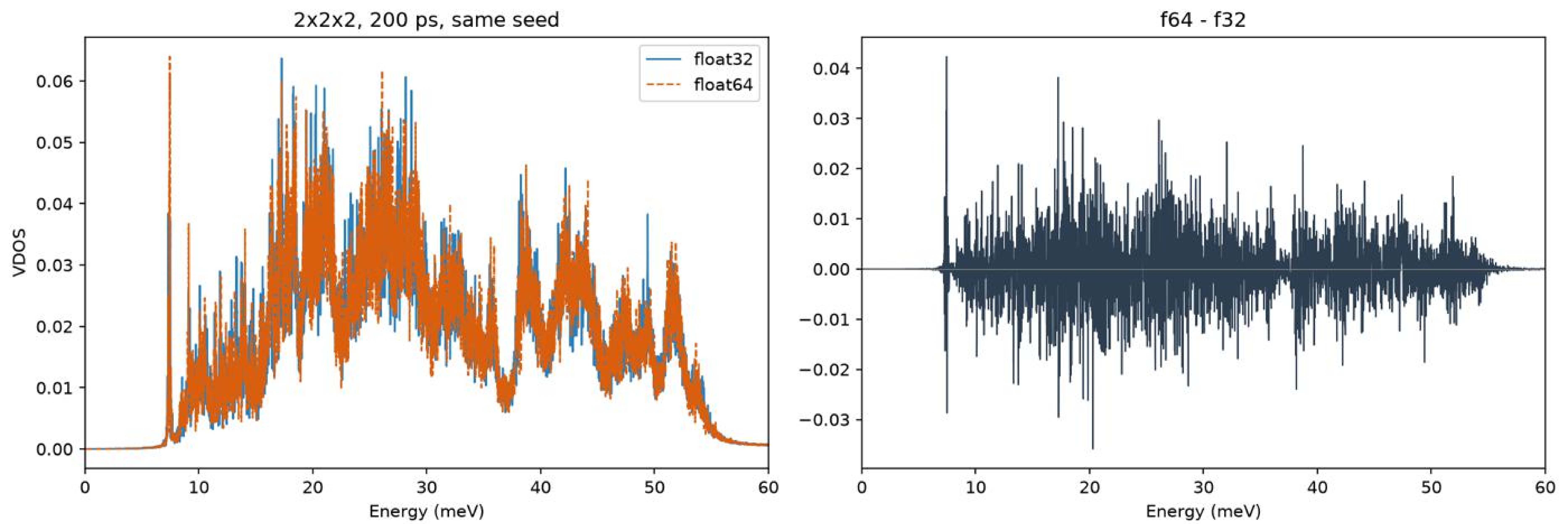
O1: MACE vs MatterSim on the 265-atom W-AlCoNi approximant
Energy (eV/atom)
0
−1
−2
−3
−4
−5
MACE-MPA-0
MatterSim-v1-1M
lower = more bound
Volume change (%)
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
MACE-MPA-0
MatterSim-v1-1M
vs DFT structure
Max displacement (A)
1.0
0.8
0.6
0.4
0.2
0.0
MACE-MPA-0
MatterSim-v1-1M
atomic rearrangement
Min interatomic dist (A)
2.0
1.5
1.0
0.5
0.0
MACE-MPA-0
MatterSim-v1-1M
after 300 K MD



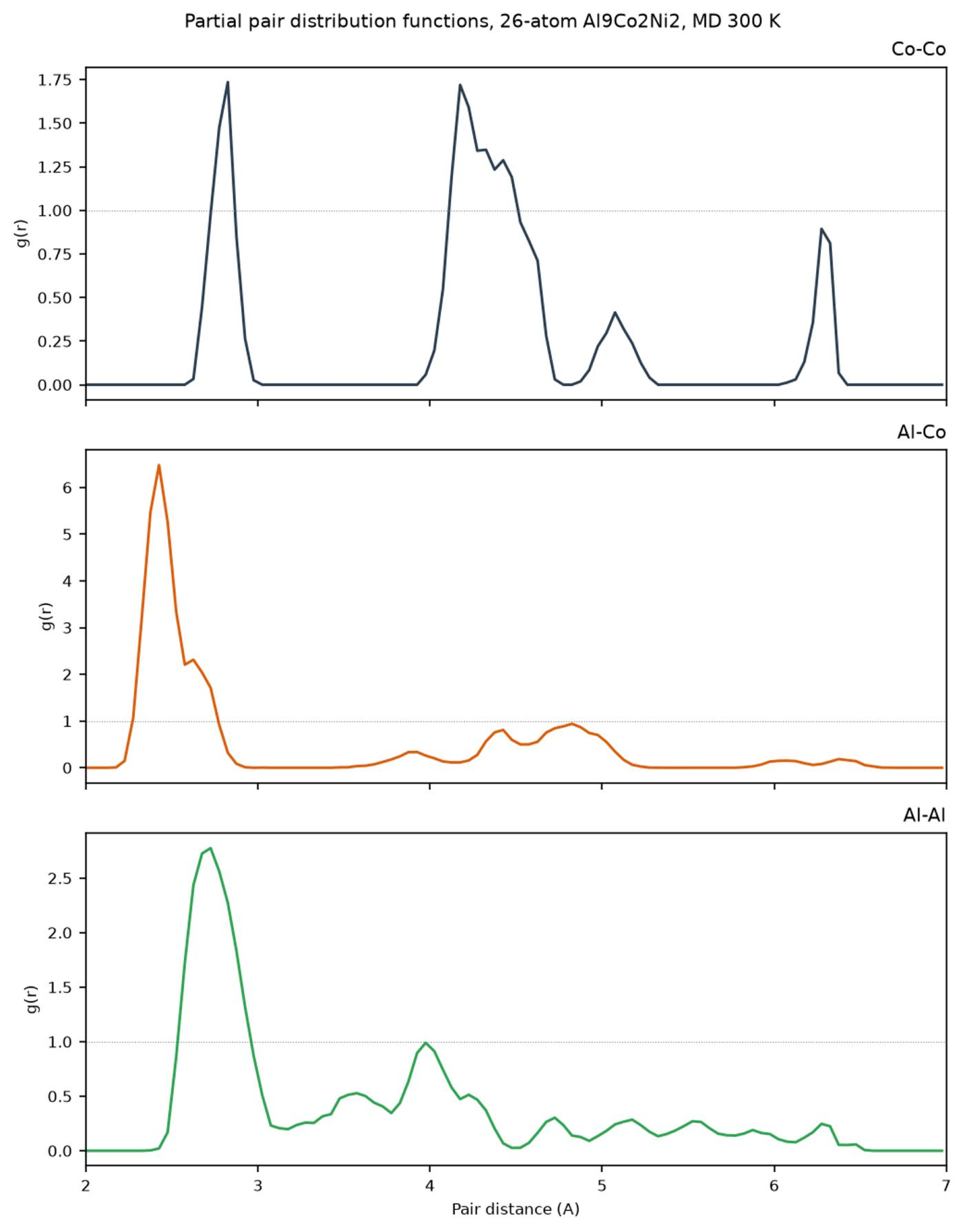
O1: MACE stability across size series, plus model comparison on W-AlCoNi
Relaxation: max force (MACE)
0.01 target
eV/A
0.000
0.002
0.004
0.006
0.008
0.010
Al9Co2Ni2 (26)
Al2CoNi (60)
W-AlCoNi (265)
MD stability: min distance (MACE)
collapse threshold
A
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Al9Co2Ni2 (26)
Al2CoNi (60)
W-AlCoNi (265)
W-phase: MACE vs MatterSim
vol change (%)
max disp (A)
−1.0
−0.5
0.0
0.5
1.0
MACE
MatterSim



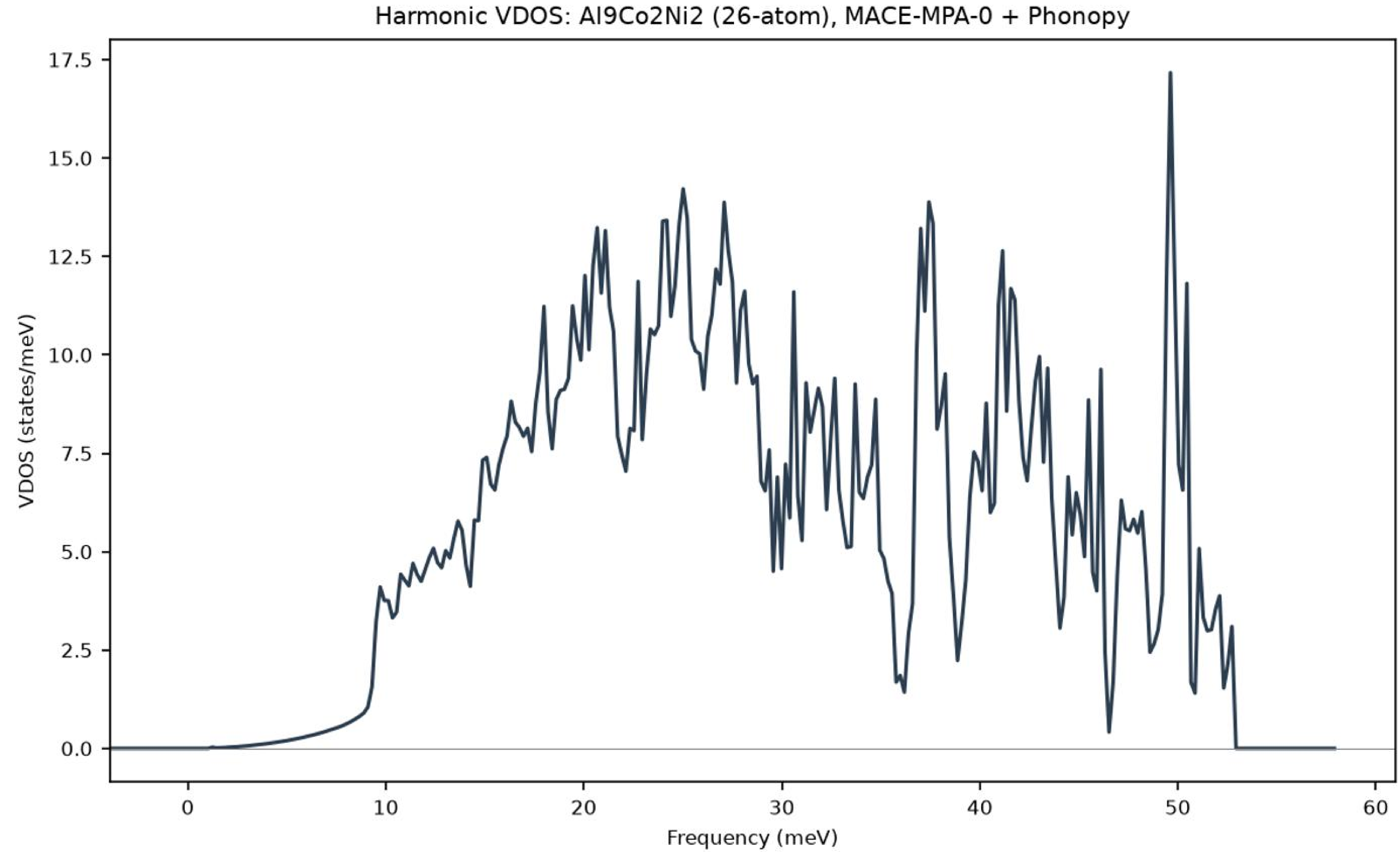
Low-frequency VDOS vs Debye prediction (Suck & Mihalkovic comparison)
26-atom Al9Co2Ni2
60-atom Al2CoNi
MACE harmonic VDOS
Debye ω^2 fit
VDOS (states/meV)
Frequency (meV)

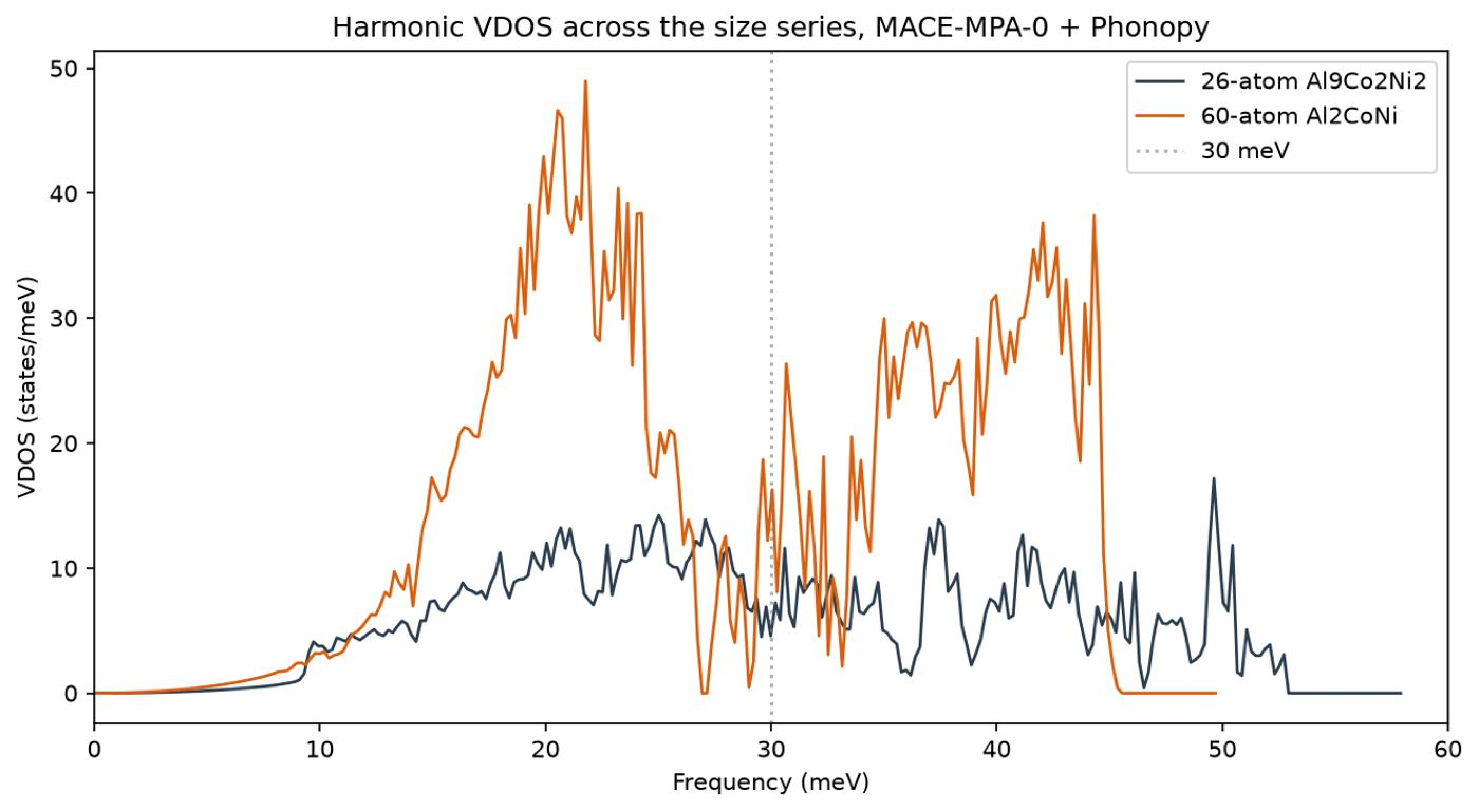


2x2x2, 200 ps, same seed
float32
float64
VDOS
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0
10
20
30
40
50
60
Energy (meV)
f64 - f32
0.04
0.03
0.02
0.01
0.00
−0.01
−0.02
−0.03
0
10
20
30
40
50
60
Energy (meV)

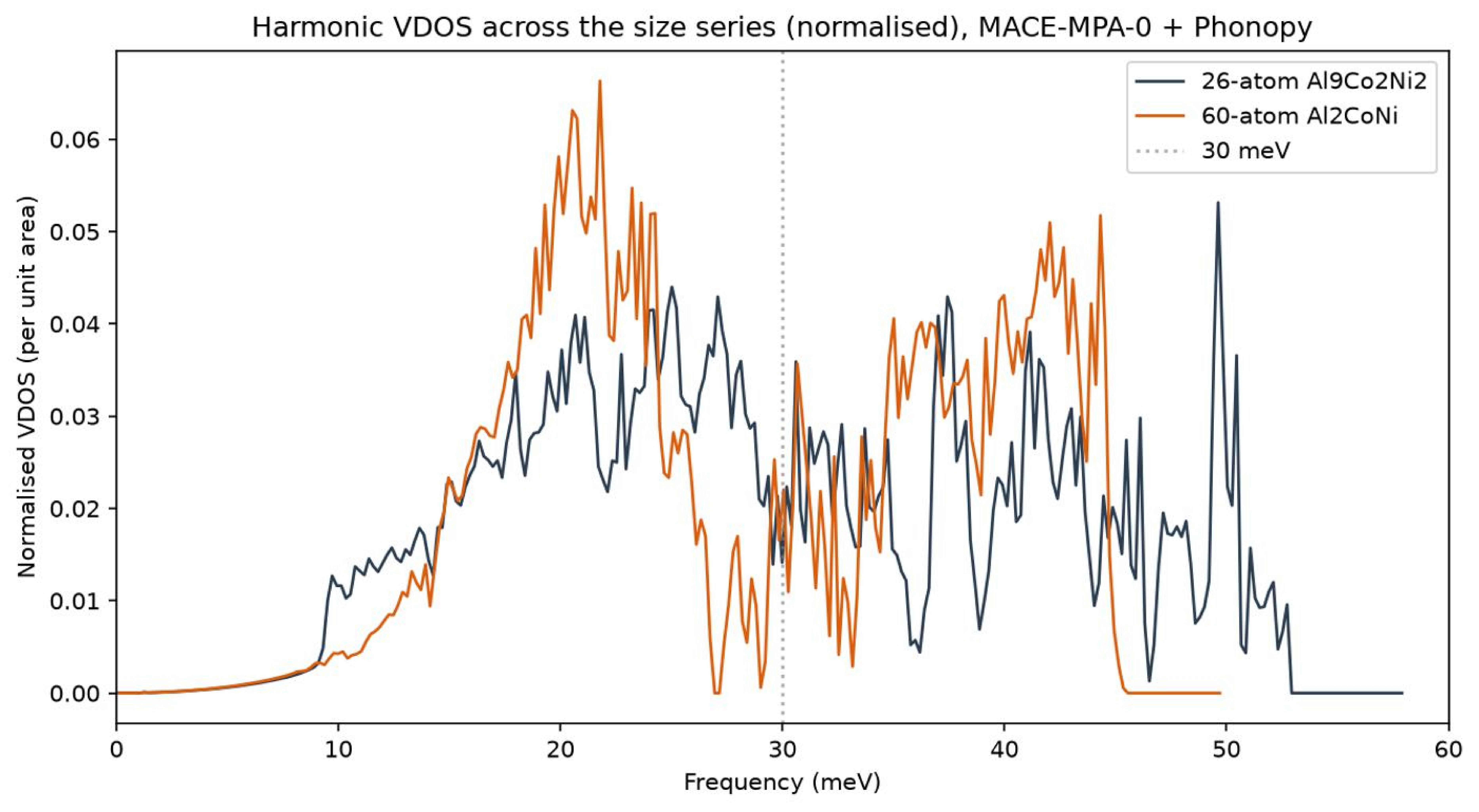


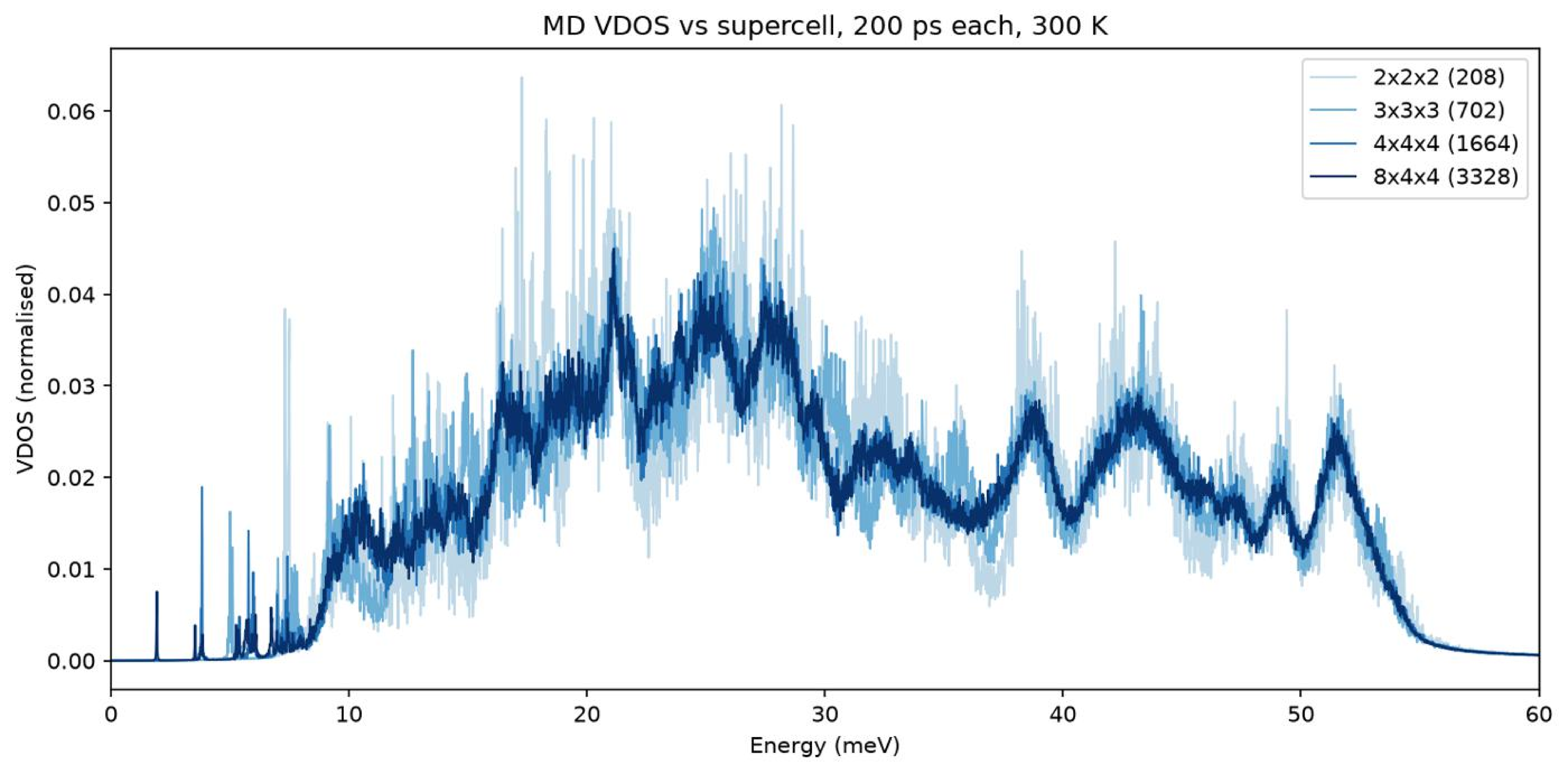
Partial pair distribution functions, 26-atom Al9Co2Ni2, MD 300 K
Co-Co
Al-Co
Al-Al
g(r)
Pair distance (A)

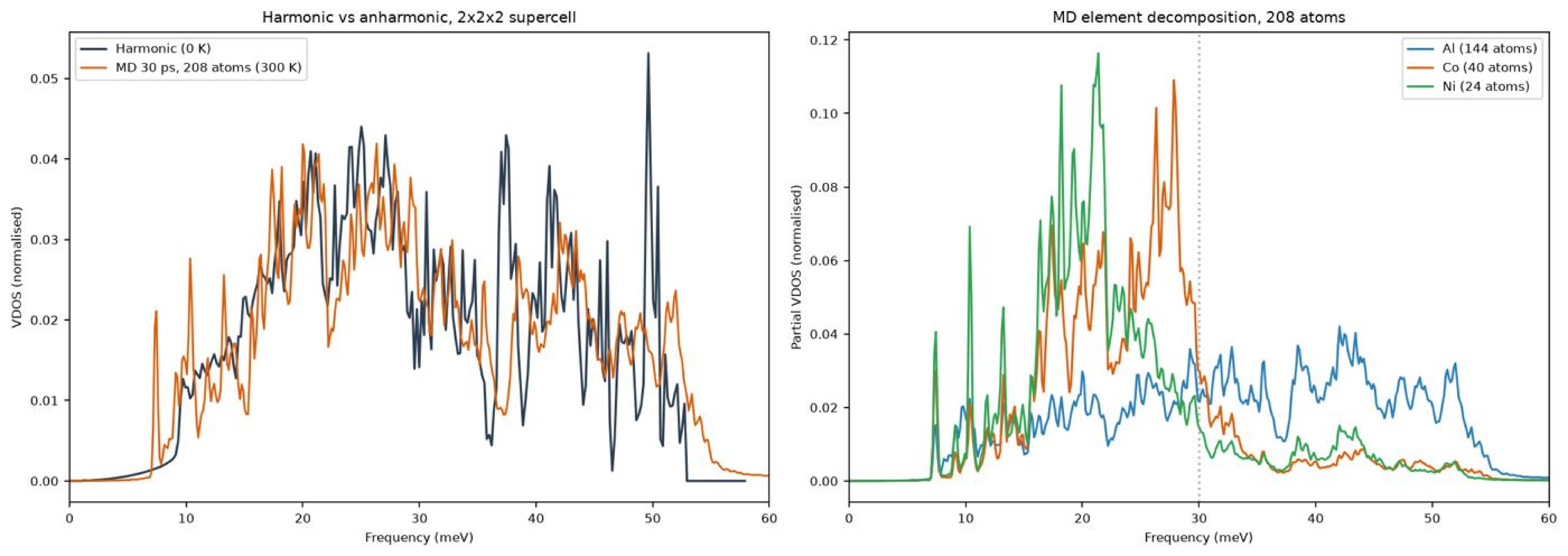

Harmonic VDOS: Al9Co2Ni2 (26-atom), MACE-MPA-0 + Phonopy
VDOS (states/meV)
Frequency (meV)
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
0
10
20
30
40
50
60



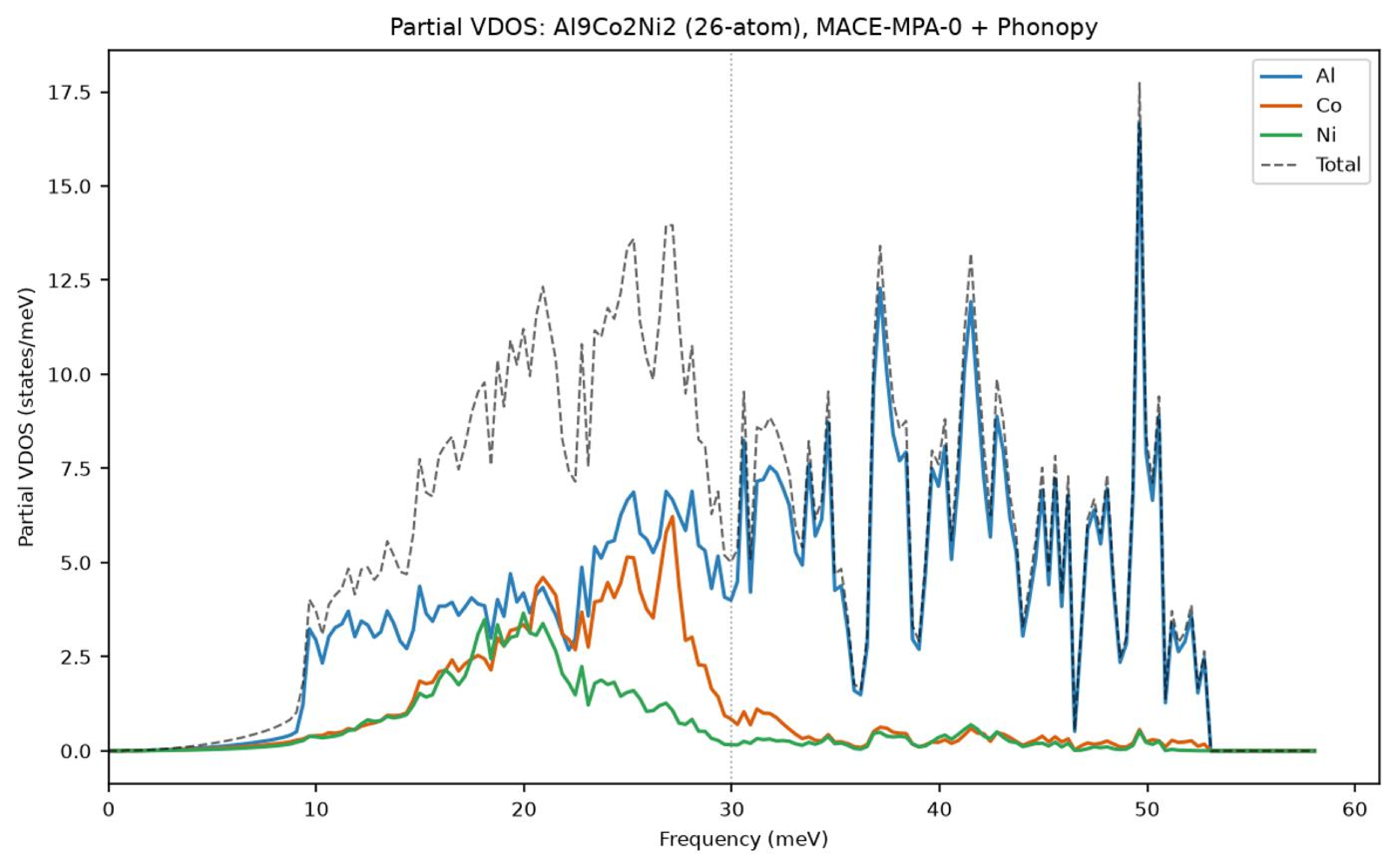
Harmonic VDOS across the size series, MACE-MPA-0 + Phonopy
VDOS (states/meV)
Frequency (meV)
26-atom Al9Co2Ni2
60-atom Al2CoNi
30 meV
0
10
20
30
40
50
60


Harmonic VDOS across the size series (normalised), MACE-MPA-0 + Phonopy
26-atom Al9Co2Ni2
60-atom Al2CoNi
30 meV
Normalised VDOS (per unit area)
Frequency (meV)
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0
10
20
30
40
50
60

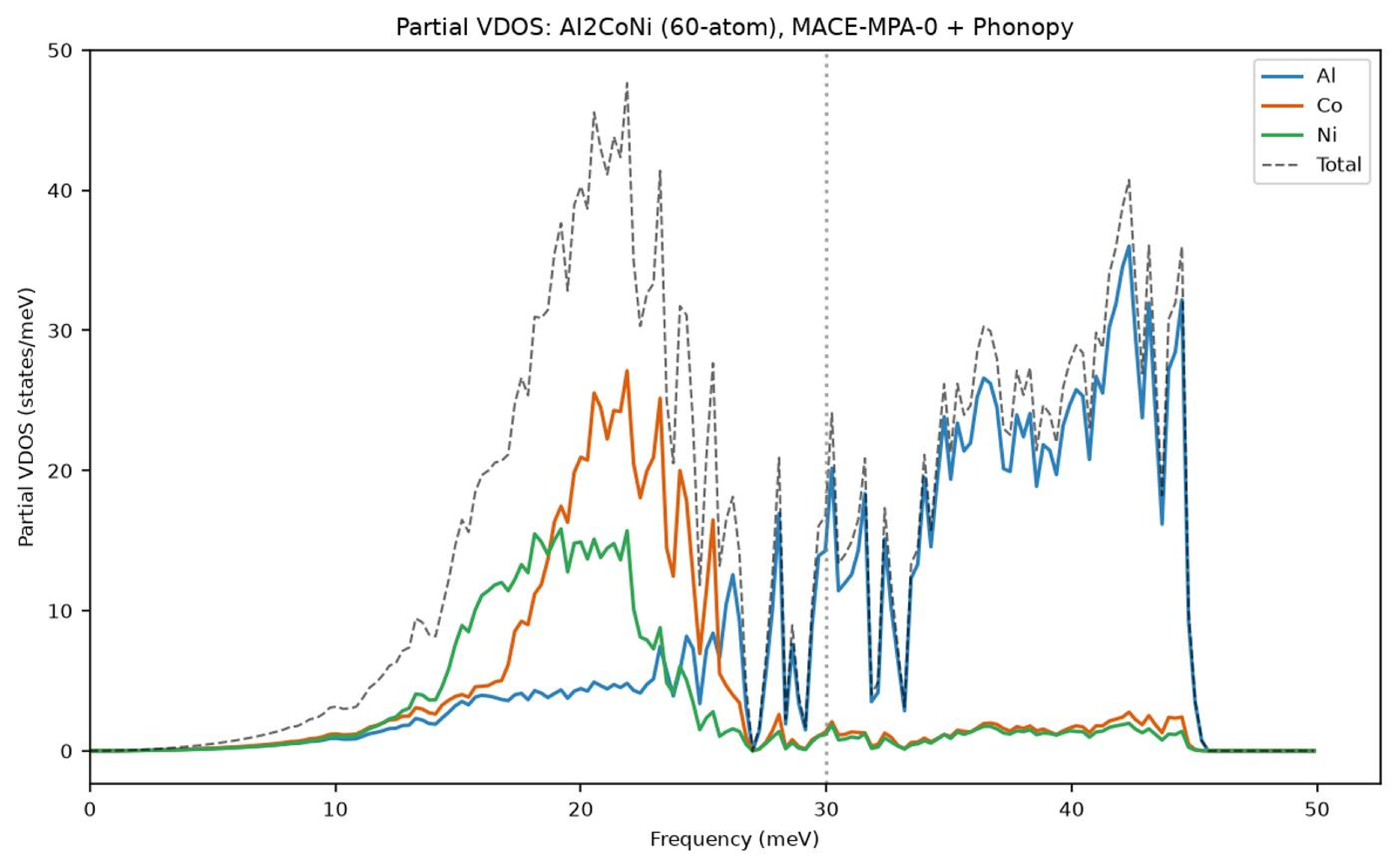

MD VDOS vs supercell, 200 ps each, 300 K
2x2x2 (208)
3x3x3 (702)
4x4x4 (1664)
8x4x4 (3328)
VDOS (normalised)
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0
10
20
30
40
50
60
Energy (meV)



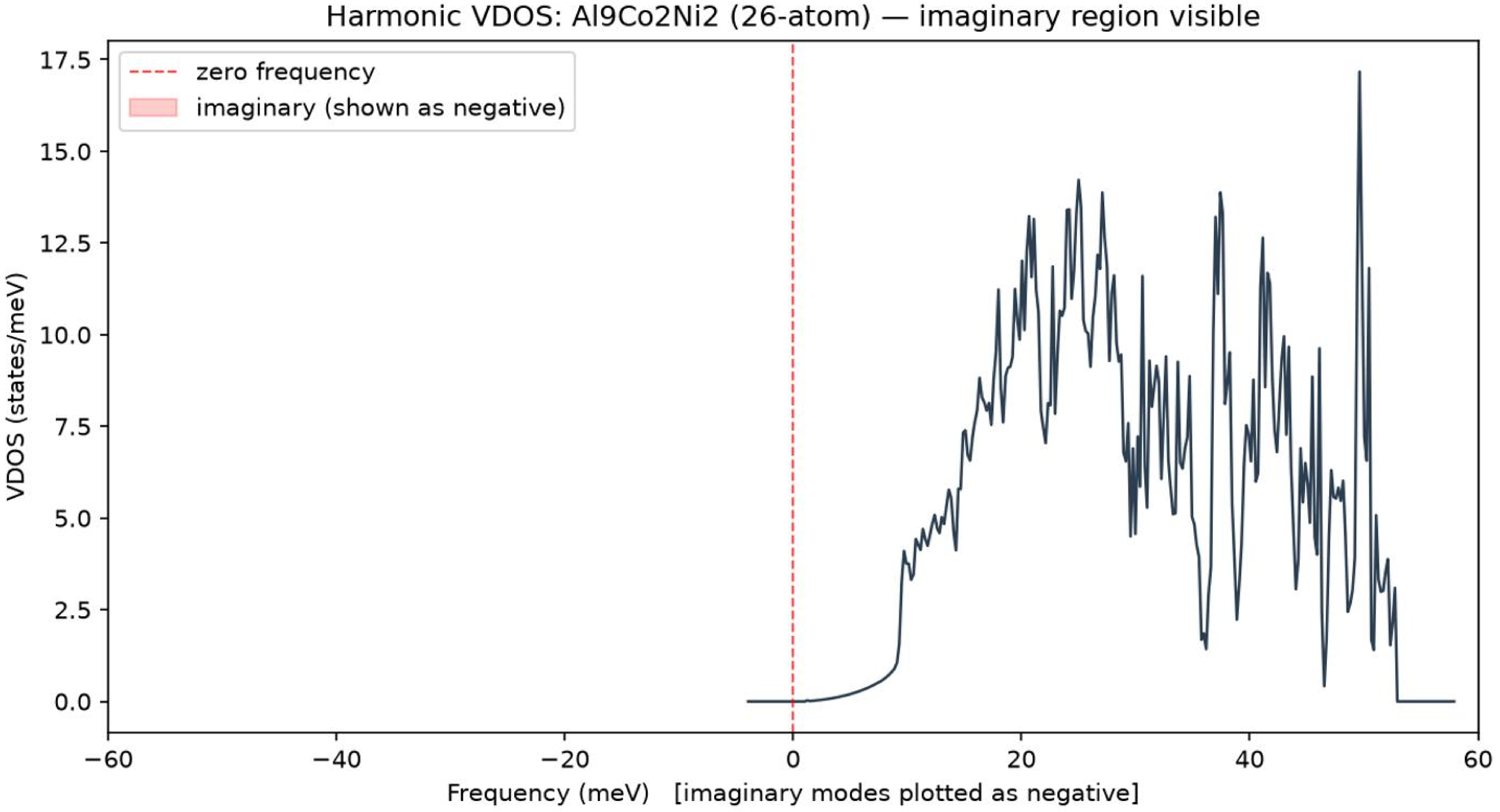
Harmonic vs anharmonic, 2x2x2 supercell
Harmonic (0 K)
MD 30 ps, 208 atoms (300 K)
VDOS (normalised)
Frequency (meV)
MD element decomposition, 208 atoms
Al (144 atoms)
Co (40 atoms)
Ni (24 atoms)
Partial VDOS (normalised)
Frequency (meV)



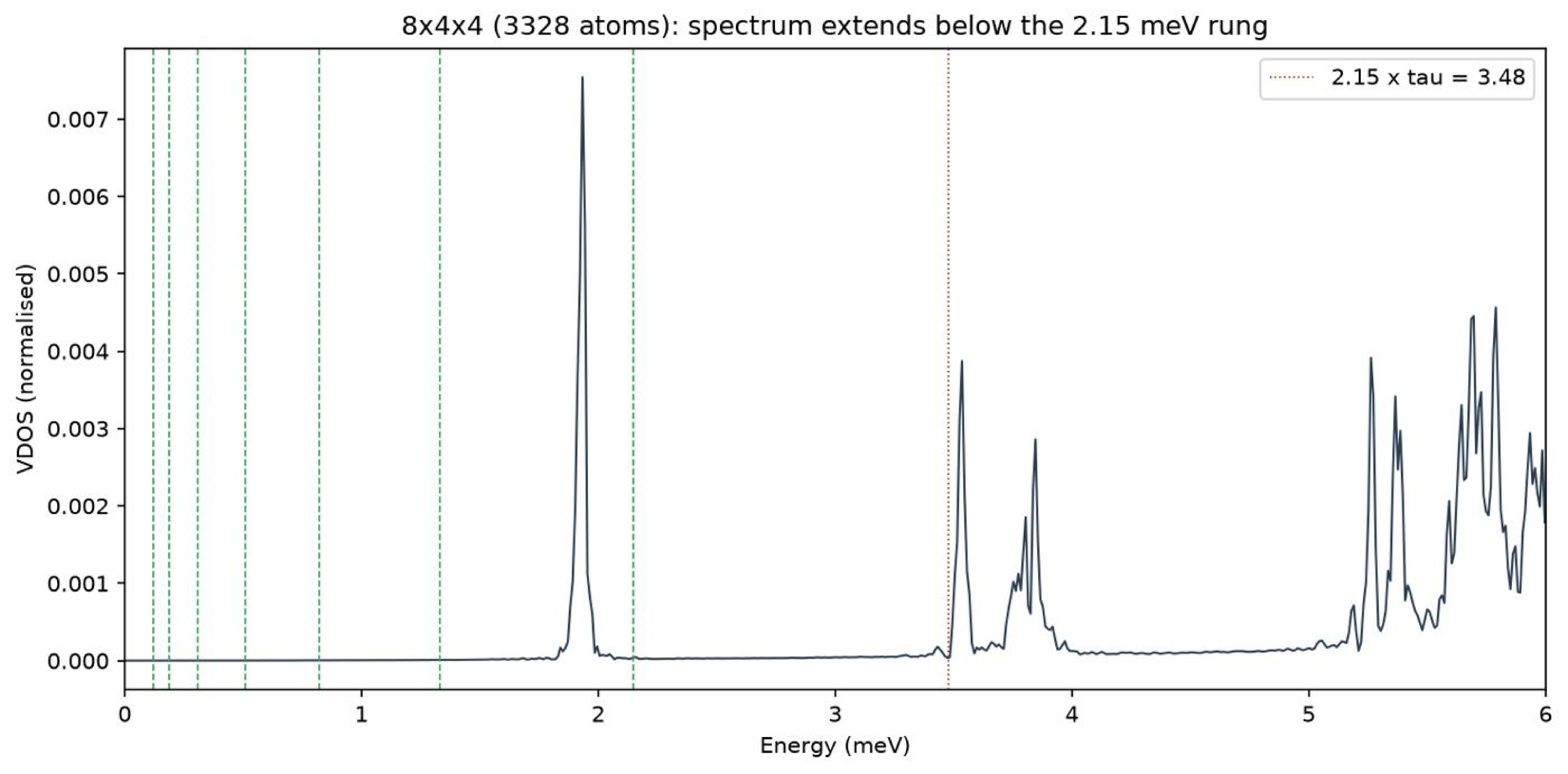
Partial VDOS: Al9Co2Ni2 (26-atom), MACE-MPA-0 + Phonopy
Al
Co
Ni
Total
Partial VDOS (states/meV)
Frequency (meV)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
10
20
30
40
50
60

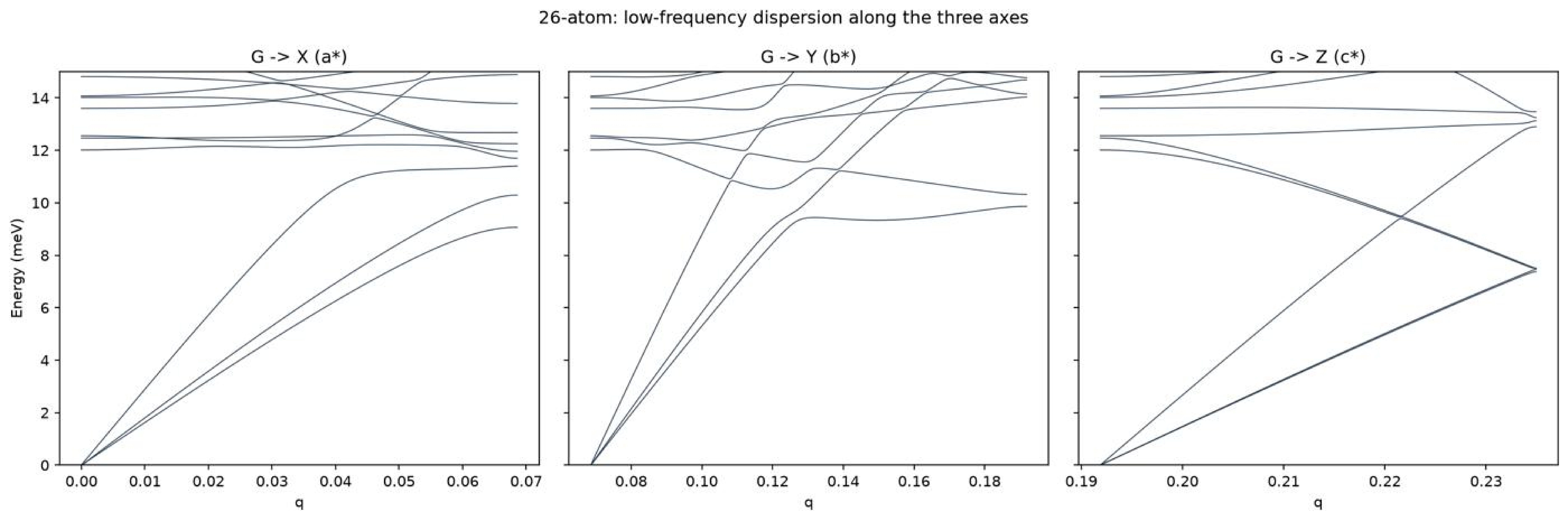


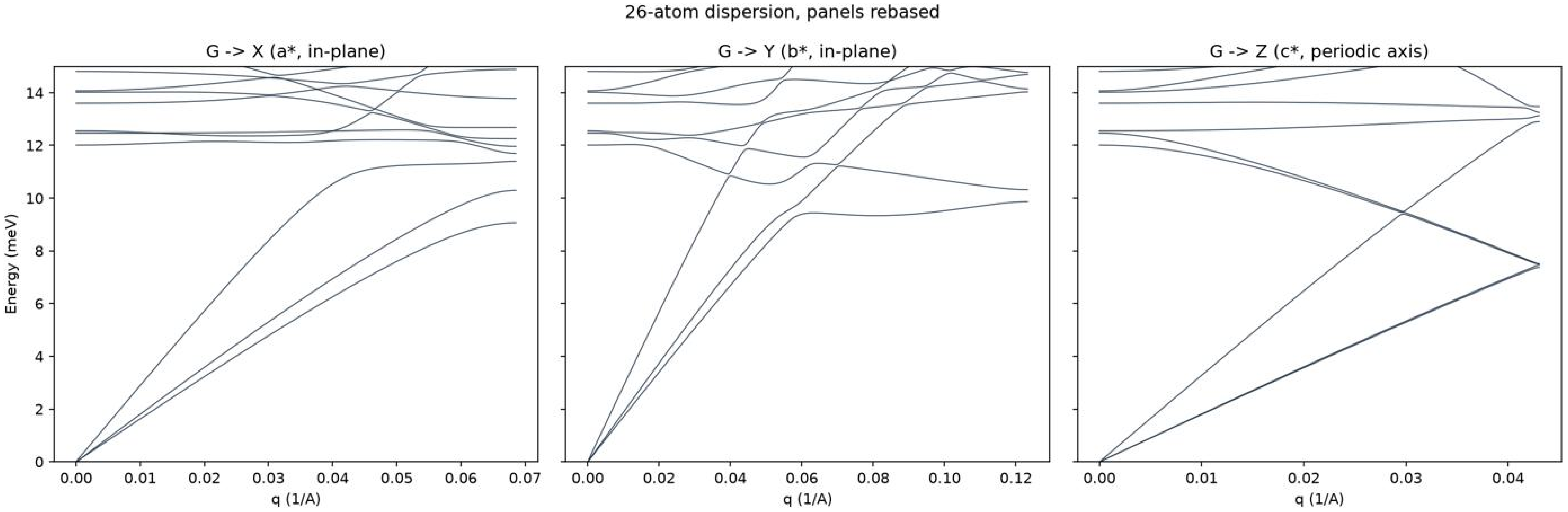
Partial VDOS: Al2CoNi (60-atom), MACE-MPA-0 + Phonopy
Al
Co
Ni
Total
Partial VDOS (states/meV)
Frequency (meV)
0
10
20
30
40
50

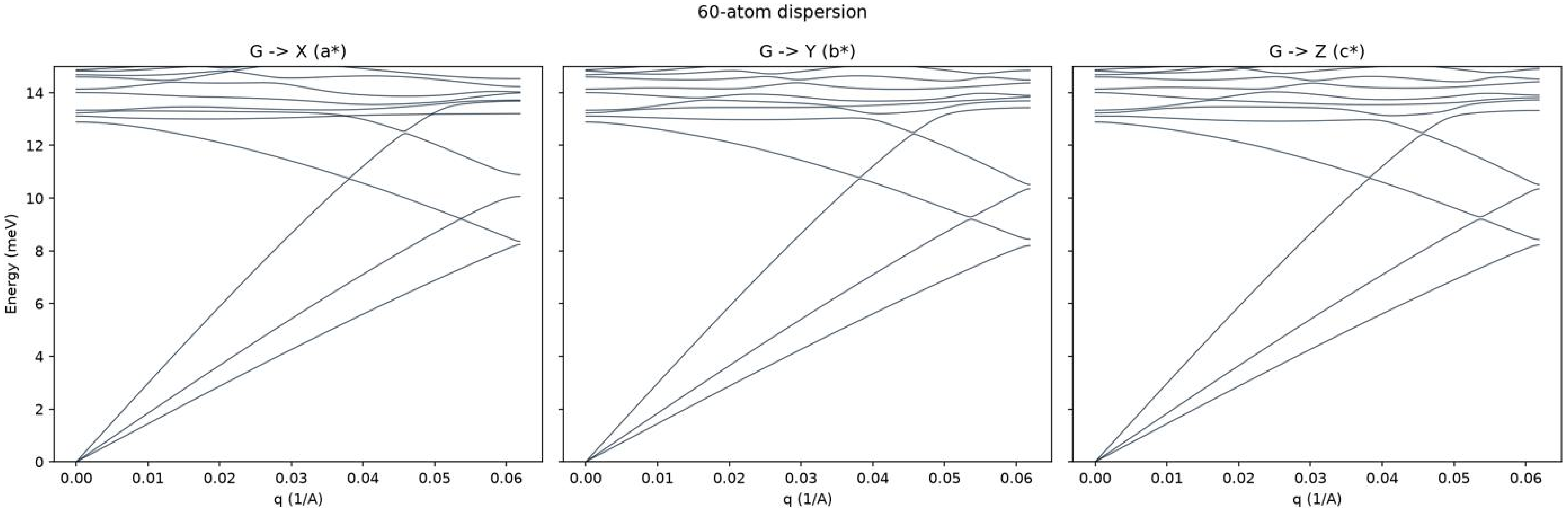


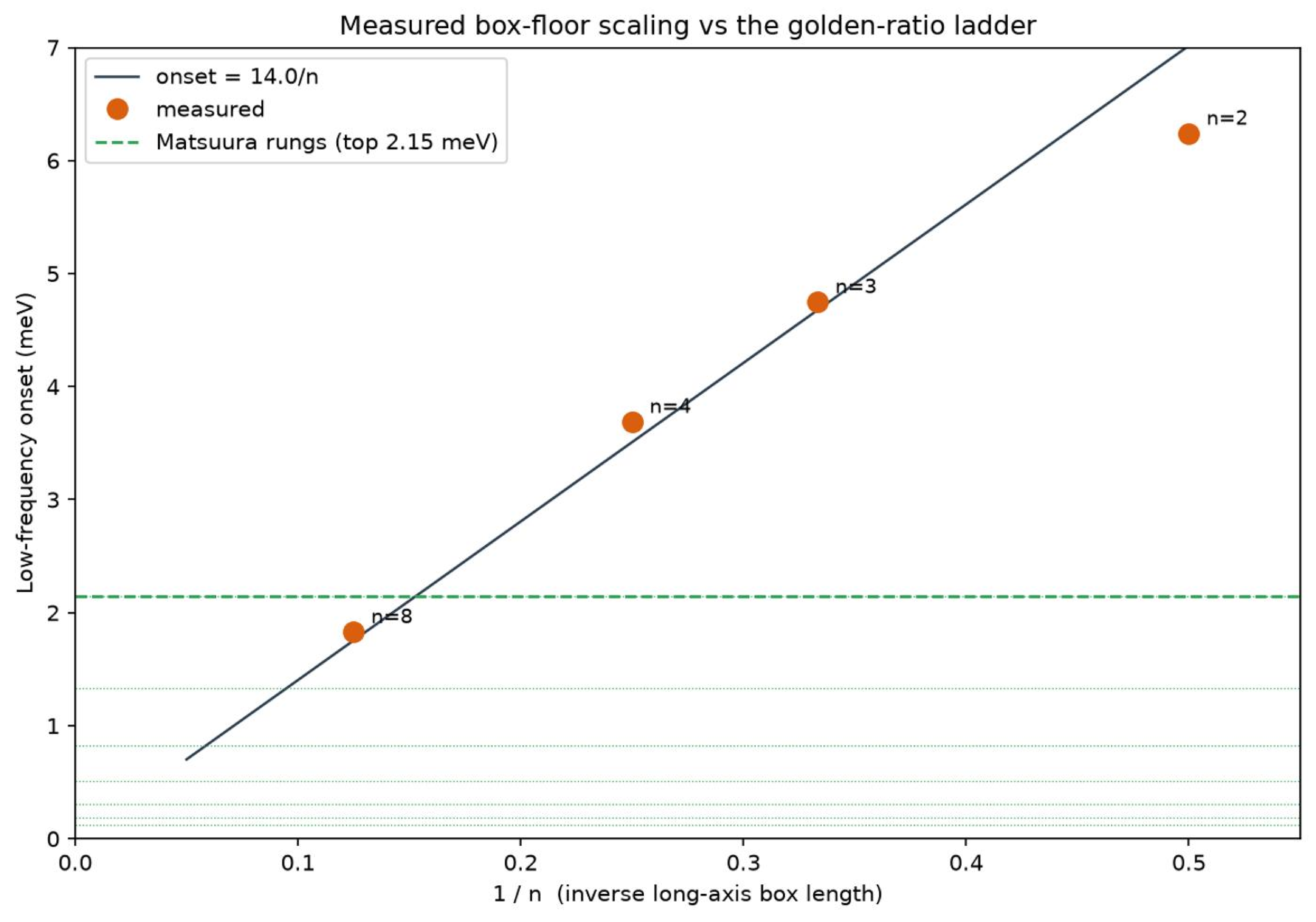
Harmonic VDOS: Al9Co2Ni2 (26-atom) — imaginary region visible
zero frequency
imaginary (shown as negative)
VDOS (states/meV)
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
−60
−40
−20
0
20
40
60
Frequency (meV)   [imaginary modes plotted as negative]


8x4x4 (3328 atoms): spectrum extends below the 2.15 meV rung
2.15 x tau = 3.48
VDOS (normalised)
0.007
0.006
0.005
0.004
0.003
0.002
0.001
0.000
0
1
2
3
4
5
6
Energy (meV)


26-atom: low-frequency dispersion along the three axes
G -> X (a*)
G -> Y (b*)
G -> Z (c*)
Energy (meV)
q
0
2
4
6
8
10
12
14
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.08
0.10
0.12
0.14
0.16
0.18
0.19
0.20
0.21
0.22
0.23


26-atom dispersion, panels rebased
G -> X (a*, in-plane)
G -> Y (b*, in-plane)
G -> Z (c*, periodic axis)
Energy (meV)
q (1/A)
q (1/A)
q (1/A)
0
2
4
6
8
10
12
14
0.00
0.01
0.02
0.03
0.04
0.05
0.06
0.07
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.00
0.01
0.02
0.03
0.04


60-atom dispersion
G -> X (a*)
G -> Y (b*)
G -> Z (c*)
Energy (meV)
q (1/A)
0
2
4
6
8
10
12
14
0.00
0.01
0.02
0.03
0.04
0.05
0.06


Measured box-floor scaling vs the golden-ratio ladder
onset = 14.0/n
measured
Matsuura rungs (top 2.15 meV)
n=2
n=3
n=4
n=8
Low-frequency onset (meV)
1 / n (inverse long-axis box length)
0
1
2
3
4
5
6
7
0.0
0.1
0.2
0.3
0.4
0.5